Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

July 10, 2026

1 Roadmap and rigour status

This lab-book is the *paper-track* of the project: it reproduces the af-validated (T0) results of the registry in `argument/` together with a thin spine of open conjectures that orients them. It is deliberately *not* a mirror of the whole registry — the non-rigorous results (inherited `proved-mod-audit`, `conjecture`, `numerical`, `heuristic`) are anchored once, honestly, in the status ledger of §21 and are never restated as theorems here. The north star is `op-classical`; its status is *open*, and its status-ledger anchor is 1. The definitions used below are the canonical shards `def-stochastic`, `def-almost-idempotent`, `def-near-positive-projection`, `def-signed-idempotent`, `def-negative-mass`, `def-height`, `def-visible-set`, `def-invisible-mass`, and `def-exposed`; this report references those identifiers and introduces no parallel definitions.

The target itself is the classical (stochastic) stability statement: there are universal constants $\eta_0, C > 0$ such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta \le \eta_0$ admits a stochastic idempotent E with

$$\|Q - E\|_{\infty\to\infty} \;\le\; C\sqrt{\eta}, \tag{1}$$

with the exponent $\frac{1}{2}$ sharp. This remains **open** in this repo.

The rigorous line is narrow by design. A result may be called rigorous here only when it is byte-matched to a local theorem under `refs/`, validated in an `af` workspace, or Lean-proved. At the time of this shard the registry records **29 af-validated results**. Those reproduced as full sections of this report are the signed–stochastic bridge 1; the halo-collapse family 2, 3, 4, 5, 6, 22, 23, 24, 25; the (EX)/factorization and fan-payment chain 7, 8, 9, 10, 11, 12; the argmin engine 13; the hiddenness witnesses and their consequences 14, 15, 16, 17, 18; the slab/capacity primitives 19, 20, 21; and the starvation obstruction 26. Every other registry result remains anchored only in the status ledger, with its lower-rung status left intact.

The working headline is itself status-sensitive. The inherited campaign distinguishes the realizable-family *linear law* from the worst-case $\sqrt{\cdot}$*-envelope*: along the realizable family the tight relation between negative mass δ and height H is

$$\delta = \tfrac{1}{2}\,H \qquad \text{(linear)},$$

while the quadratic form $\delta \gtrsim H^2$ is only the worst-case envelope, binding because $H \le O(\tau) = O(\sqrt{\delta})$. The $\sqrt{\eta}$ exponent of (1) is nonetheless sharp, certified by the family `ex-hume` (anchored at 1), which is still `proved-mod-audit` here rather than one of the rigorous results. Crossing between the signed and stochastic pictures should cite the validated bridge 1 rather than re-normalising notation locally.

2 The signed-stochastic bridge

Lemma 1 (`lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal $C > 0$ such that:*

1. *(stochastic $\to$ signed) if Q is row-stochastic with $\|Q^2 - Q\|_{\infty\to\infty} \le \eta$, then $P = \theta(2Q - 1)$ is a signed affine retraction (`def-signed-idempotent`) satisfying $\|P - Q\|_{\infty\to\infty} \le C\eta$ and negative mass $\delta \le C\eta$;*

2. *(signed $\to$ stochastic) conversely, row-normalising the positive parts p_i^+ of a signed idempotent P yields a row-stochastic Q with $\|P - Q\|_{\infty\to\infty} \le 2\delta$ and $\|Q^2 - Q\|_{\infty\to\infty} \le 6\delta + 4\delta^2$.*

Transcription note. *The contract writes the map norm as $\|\cdot\|$ without a subscript; per `CONVENTIONS.md` (b) it is the $\infty \to \infty$ map norm $\|\cdot\|_{\infty\to\infty}$. The operator θ (the signed retraction $\theta(2Q - 1)$) is carried verbatim from the contract, which does not expand it.*

Registry contract (verbatim).

```
The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent
up to universal constants: Q row-stochastic with ||Q^2-Q|| <= eta gives P=theta(2Q-1) signed affine
retraction with ||P-Q|| <= C eta and neg mass delta <= C eta, and conversely row-normalising p_i^+
gives Q with ||P-Q|| <= 2 delta, ||Q^2-Q|| <= 6 delta+4 delta^2.
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 29-node adversarial tree with root `validated` and clean taint. The human-readable export is `proofs/lem-classical-equiv/export.md`; the export also records the earlier missing-definition challenge and its repair by importing `def-negative-mass`.
Role in the chain. This bridge is the authorized way to move between `op-classical`'s stochastic formulation and the signed exact-idempotent registry. It is a dependency for the inherited simplex, rank-one, and approximate-simplex reductions, but those downstream reductions remain `proved-mod-audit` unless and until they clear an L0 gate.

3 Height collapse

Observation 2 (`obs-height-collapse`)**.** *Let P be an exact signed idempotent with $0 < \delta(P) \leq \frac{1}{4}$, nonempty visible set $W(P)$, and a hidden top vertex v of height $H = \mathrm{dist}_1(p_v, \mathrm{conv}\, W)$ maximal among the rows. Then the invisible mass σ_v and the row negative mass ν_v satisfy*

$$H\,(1 - \sigma_v) \;\leq\; \nu_v\,(2 + 4\delta).$$

Registry contract (verbatim).

```
Height collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty visible set
W(P), and hidden top vertex v (height H = dist_1(p_v, conv W) maximal among rows), the invisible
mass sigma_v and the row negative mass nu_v satisfy H * (1 - sigma_v) <= nu_v * (2 + 4*delta).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 19-node adversarial tree with root `validated` and taint `19/19 clean`. The export is `proofs/obs-height-collapse/export.md`. Earlier compound "hence" clauses were excluded from the contract before validation; this section records only the validated inequality.
Role in the chain. This result explains why invisible mass alone does not finish the Kernel/`(EX)` route: height is controlled only after pricing the complementary mass. A future dimension-free cap on the relevant invisible mass would combine with this inequality, but that cap is not proved here.

4 Mass split bookkeeping

Lemma 3 (`lem-mass-split`). *Let P be an exact signed idempotent and v any row index. Writing $a_j = P_{vj}$, $a_j^+ = \max(a_j, 0)$, $a_j^- = \max(-a_j, 0)$, and $\nu_v = \sum_j a_j^-$, one has*

$$\sum_j a_j^+ \; = \; 1 + \nu_v.$$

Registry contract (verbatim).

```
Mass split: for an exact signed idempotent P and any row index v, writing a_j = P_{vj}, a_j^+ =
max(a_j, 0), a_j^- = max(-a_j, 0), and nu_v = sum_j a_j^-, one has sum_j a_j^+ = 1 + nu_v.
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a clean 9-node adversarial tree with root `validated` and taint `9/9 clean`. The export is `proofs/lem-mass-split/export.md`.
Role in the chain. This is the mass bookkeeping dependency factored out of the first `lem-halo-collapse` elevation attempt. Its purpose is to keep later halo-collapse proof trees from re-deriving the same row-sum identity in multiple sibling branches.

5 Convex outsourcing

Lemma 4 (`lem-residual-lower`). *Let* $C = \operatorname{conv}\{y_1, \dots, y_N\} \subseteq \mathbb{R}^n$ *be the convex hull of finitely many points, and suppose*

$$p = \sum_{j=1}^{m} c_j \, p_j + (1 - s) \, q, \qquad c_j \geq 0, \quad s = \sum_{j=1}^{m} c_j < 1,$$

with $p_j, q \in \mathbb{R}^n$ *and* $\operatorname{dist}_1(p_j, C) \leq \operatorname{dist}_1(p, C)$ *for every* j*. Then*

$$\operatorname{dist}_1(p, C) \; \leq \; \operatorname{dist}_1(q, C).$$

Registry contract (verbatim).

```
Convex outsourcing: let C be the convex hull of finitely many points of R^n, and suppose p =
sum_{j=1}^m c_j p_j + (1 - s) q with points p_j, q in R^n, coefficients c_j >= 0, s = sum_{j=1}^m
c_j < 1, and dist_1(p_j, C) <= dist_1(p, C) for every j; then dist_1(p, C) <= dist_1(q, C).
```

Status. `proved`; `af:  validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records 32 validated live nodes, 36 total nodes including 4 archived nodes, root `validated`, and taint `36/36 clean`. The export is `proofs/lem-residual-lower/export.md`.
Role in the chain. This is a frame-free convex geometry dependency for `lem-halo-collapse`. It supplies the lower residual-distance comparison used when a hidden row is split into ordinary mass plus a residual term.

6 Residual distance bound

Lemma 5 (`lem-residual-upper`). *Let $C \subseteq \mathbb{R}^n$ be the convex hull of finitely many points, let $b_1, \dots, b_M \geq 0$ and $c_1, \dots, c_N \geq 0$ with $m = \sum_j b_j - \sum_k c_k > 0$, let $p_j, r_k \in \mathbb{R}^n$ with*

$$q = \frac{\sum_j b_j\, p_j - \sum_k c_k\, r_k}{m},$$

and let $D_k \geq 0$ satisfy $\|x - r_k\|_1 \leq D_k$ for all $x \in C$ and each k. Then

$$m\ \mathrm{dist}_1(q, C)\ \leq\ \sum_j b_j\ \mathrm{dist}_1(p_j, C) + \sum_k c_k\, D_k.$$

Registry contract (verbatim).

```
Residual distance bound: let C be the convex hull of finitely many points of R^n, let b_1..b_M,
c_1..c_N >= 0 with m = sum_j b_j - sum_k c_k > 0, let p_j, r_k be points of R^n with q = (sum_j b_j
p_j - sum_k c_k r_k) / m, and let D_k >= 0 satisfy ||x - r_k||_1 <= D_k for all x in C and each k;
then m * dist_1(q, C) <= sum_j b_j * dist_1(p_j, C) + sum_k c_k * D_k.
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records 49 validated live nodes, 52 total nodes including 3 archived nodes, root `validated`, and taint `52/52 clean`. The export is `proofs/lem-residual-upper/export.md`. The tree size is above the 12-node brittleness target, so the tracked refactor warning remains honest rather than being hidden in prose.
Role in the chain. This is the companion frame-free convex geometry dependency for `lem-halo-collapse`. Together with 3 and 4, it packages the residual-recombination estimate needed by the halo-collapse bridge, now itself validated in 6.

7 Halo-robust height collapse

Lemma 6 (`lem-halo-collapse`). *Let P be an exact signed idempotent with $0 < \delta(P) \leq \frac{1}{4}$, nonempty visible set $W(P)$, and hidden top vertex v of height H. Let σ be the invisible mass of v, let σ_g be the halo-robust invisible mass—the positive-coefficient mass v places on rows at ℓ_1 distance $> \tau/4$ from* $\operatorname{conv} W$*, where $\tau = \sqrt{\delta}$—and let ν_v be the row negative mass. Then*

$$H\,(1-\sigma_g) \;\leq\; (\sigma-\sigma_g)\,\tfrac{\tau}{4} + \nu_v\,(2+4\delta).$$

Registry contract (verbatim).

```
Halo-robust height collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty
visible set W(P), and hidden top vertex v of height H, let sigma be the invisible mass of v,
sigma_g the halo-robust invisible mass (the positive coefficient mass v places on rows at ell-1
distance > tau/4 from conv W, tau = sqrt(delta)), and nu_v the row negative mass; then H * (1 -
sigma_g) <= (sigma - sigma_g) * tau/4 + nu_v * (2 + 4*delta).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 20-node adversarial tree, root `validated`, taint `20/20 clean` (run 2 on the factored workspace). The export is `proofs/conj-halo-collapse/export.md`. The three `af`-validated dependencies 3, 4, and 5 are imported as externals, so the root rests only on validated or cited inputs.
Elevation history. Run 1 (pre-factoring) ballooned past the node cap (49 live nodes over the cap of 40) and was aborted as a structure signal, not a mathematical refutation: the mass-split bookkeeping and the residual-distance estimates were being re-derived inline across siblings. Factoring those into the three registry sub-lemmas above cured it; run 2 closed at 20 nodes.
Role in the chain. This bridge refines the validated collapse bound 2 by pricing recipients inside the $\tau/4$-halo of the visible hull at their actual distance rather than the worst case H, so the bound stays non-vacuous even when the raw invisible mass exceeds one through self or halo mass (1). Together with a halo-robust cap $\tilde{\sigma}_g \leq 1-c$ for hidden top vertices it would give $H = O(\tau)$, the height cap of the Kernel Conjecture 1. That cap is OPEN: nothing in this section promotes the kernel chain.

8 Factorization bound

Lemma 7 (`lem-factorization`)**.** *Let P be an exact signed idempotent (a square real matrix with $P^2 = P$ and all row sums equal to 1), let $\delta(P) = \max_i \sum_j \max(-P_{ij}, 0)$, let $k = \operatorname{rank} P$, and let $U = (u_1, \ldots, u_k)$ be an actual-row basis of P—the rows $p_{u_1}, \ldots, p_{u_k}$ form a basis of the row space—with Gram volume $\mathrm{Vol}(U) \geq \frac{1}{2}\,\mathrm{Vol}_{\max}(P)$, where $\mathrm{Vol}_{\max}(P)$ is the maximum Gram volume over all actual-row bases of P. Define coordinates $a_t(j)$ by $p_j = \sum_t a_t(j)\, p_{u_t}$, and for each pivot index $s \in \{1, \ldots, k\}$ set*

$$\beta_s(j) = P_{u_s j}, \qquad \lambda_s(j) = 1 - a_s(j), \qquad \mu_s(j) = \sum_{t \neq s} \max(-a_t(j), 0),$$

$$\sigma_s(j) = \sum_{t \neq s} \max(a_t(j), 0), \qquad E_s(j) = \max(\mu_s(j) - \lambda_s(j), 0),$$

$$\Phi_s(U) = \sum_j \max(\beta_s(j), 0)\, E_s(j), \qquad S_s^*(U) = \sum_j \max(\beta_s(j), 0)\big(\sigma_s(j) + 2\max(-\lambda_s(j), 0)\big).$$

Then for every pivot s,

$$S_s^*(U) \;\leq\; 2\,\Phi_s(U) + 6\,\delta(P).$$

Registry contract (verbatim).

```
Factorization bound: let P be an exact signed idempotent (square real matrix with P^2 = P and all
row sums equal to 1), let delta(P) = max_i sum_j max(-P_ij, 0), let k = rank(P), and let U = (u_1,
..., u_k) be an actual-row basis of P (the rows p_{u_1}, ..., p_{u_k} of P form a basis of the row
space of P) whose Gram volume satisfies Vol(U) >= (1/2) * Vol_max(P), where Vol_max(P) is the
maximum Gram volume over all actual-row bases of P; define coordinates a_t(j) by p_j = sum_t a_t(j)
p_{u_t}, and for each pivot index s in {1, ..., k} set beta_s(j) = P_{u_s j}, lambda_s(j) = 1 -
a_s(j), mu_s(j) = sum_{t != s} max(-a_t(j), 0), sigma_s(j) = sum_{t != s} max(a_t(j), 0), E_s(j) =
max(mu_s(j) - lambda_s(j), 0), Phi_s(U) = sum_j max(beta_s(j), 0) * E_s(j), and S*_s(U) = sum_j
max(beta_s(j), 0) * (sigma_s(j) + 2 * max(-lambda_s(j), 0)); then for every pivot s, S*_s(U) <= 2 *
Phi_s(U) + 6 * delta(P).
```

Status. `proved`; `af:  validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records an 11-live-node adversarial tree (12 total nodes, 1 archived), root `validated`, taint `12/12 clean`, closed in run 1 after five rounds. The export is `proofs/lem-factorization/e` The contract was narrowed to the single inequality before seeding; the inherited tightness claim for the constants $(2, 6)$ was deliberately not elevated and remains `proved-mod-audit` material in the registry shard body.
Role in the chain. This is the composition link of the `(EX)` program: any universal chart bound $\max_s \Phi_s(U_0) \leq C_0\,\delta(P)$ over the $\theta = 1/2$ class now composes rigorously to the class-score constant $C_{\mathrm{sf}} = 2C_0 + 6$. The `(EX)` input itself (1) remains a conjecture: certified numerical evidence supports a plateau near $C_0 = 2$, but nothing in this section promotes it.

9 Zero-sum triangle bound

Lemma 8 (`lem-zerosum-triangle`)**.** *Let* $w, v \in \mathbb{R}^d$ *with* v *having zero coordinate sum,* $\sum_l v(l) = 0$*, and write* $n(x) = \sum_l \max(-x(l), 0)$*. Then*

$$n(w - v) \;\leq\; n(w) + n(v).$$

Registry contract (verbatim).

```
Zero-sum triangle bound: let w and v be vectors in R^d with v having coordinate sum zero (sum_l
v(l) = 0); write n(x) = sum_l max(-x(l), 0); then n(w - v) <= n(w) + n(v).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 10-node adversarial tree, root `validated`, taint `10/10 clean`, closed in run 1 after three rounds. The export is `proofs/lem-zerosum-triangle/export.md`. The run-1 MISSING-fact challenge isolated the load-bearing hypothesis: the identity $\sum_l \max(v(l), 0) = n(v)$ holds exactly because v has zero coordinate sum; for general v only the pointwise inequality survives.
Role in the chain. First factored dependency of 10: it was split out of the `lem-fan-payment` workspace after two balloon aborts forced a linear proof chain into cross-sibling re-derivations. Downstream `af` trees import it as a validated external instead of re-deriving the triangle step per node.

10 Weighted minimum bound

Lemma 9 (`lem-weighted-min`). *Let $p_1, \dots, p_m$ be positive reals with $\sum_i p_i = 1$ and let $n_1, \dots, n_m$ be real numbers. Then*

$$\min_{i \in \{1,\dots,m\}} n_i \;\le\; \sum_i p_i \, n_i.$$

Registry contract (verbatim).

```
Weighted minimum bound: let p_1, ..., p_m be positive reals with sum_i p_i = 1 and let n_1, ...,
n_m be real numbers; then min over i in {1, ..., m} of n_i <= sum_i p_i * n_i.
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records an 8-node adversarial tree, root `validated`, taint `8/8 clean` (the build run's verifier calls were blanked by a network outage and the verify phase was resumed cleanly). The export is `proofs/lem-weighted-min/export.md`.
Role in the chain. Second factored dependency of 10: the index-selection step (some support point has value at most the weighted average). Textbook-adjacent, but registered as its own shard so `af` trees can import it as a validated external instead of re-deriving it across siblings.

11 Zero-sum fan payment

Lemma 10 (`lem-fan-payment`). *Let* $(w_1, p_1), \dots, (w_m, p_m)$ *be a finite family with vectors* $w_i \in \mathbb{R}^d$ *and weights* $p_i > 0$ *satisfying* $\sum_i p_i = 1$*, such that every* w_i *has zero coordinate sum,* $\sum_l w_i(l) = 0$*, and the weighted barycenter is zero,* $\sum_i p_i w_i = 0$*. Write* $n(w) = \sum_l \max(-w(l), 0)$*. Then*

$$\min_{i^* \in \{1,\dots,m\}} \sum_i p_i\, n(w_i - w_{i^*}) \;\le\; 2 \sum_i p_i\, n(w_i).$$

Registry contract (verbatim).

```
Zero-sum fan payment: let (w_1, p_1), ..., (w_m, p_m) be a finite family with vectors w_i in R^d
and weights p_i > 0 satisfying sum_i p_i = 1, such that every w_i has coordinate sum zero (sum_l
w_i(l) = 0) and the weighted barycenter is zero (sum_i p_i w_i = 0); write n(w) = sum_l max(-w(l),
0); then min over i* in {1, ..., m} of sum_i p_i n(w_i - w_{i*}) <= 2 * sum_i p_i n(w_i).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 15-node adversarial tree, root `validated`, taint `15/15 clean`, closed in run 3 on the factored workspace; it imports the `af`-validated dependencies 8 and 9 as externals. Runs 1–2, before the factoring, balloon-aborted at 39 and 47 nodes; the factoring cured the balloon (15 versus 47). The export is `proofs/lem-fan-payment/export.md`.
Role in the chain. The all-mass fan payment lemma is the discrete mechanism behind the certified plateau-2 constant on the reduced fan families of the arm-A campaign: in a reduced edge-fan chart the anchor-pivot score is exactly the weighted average $\sum_i p_i\, n(w_i - w_*)$, so this lemma bounds the fan-template Φ/δ by 2. The registry shard body additionally records an exact refutation certificate for the over-broad variant (arbitrary w_0, no zero-sum hypothesis, ratio 5); that certificate is wave material and is not elevated here.

12 Negative-part subadditivity

Lemma 11 (`lem-negpart-subadditive`). *For all vectors $x, y \in \mathbb{R}^d$, writing $n(w) = \sum_l \max(-w(l), 0)$, one has*

$$n(x + y) \;\leq\; n(x) + n(y).$$

Registry contract (verbatim).

```
Negative-part subadditivity: for all vectors x and y in R^d, writing n(w) = sum_l max(-w(l), 0),
one has n(x + y) <= n(x) + n(y).
```

Status. `proved`; `af:  validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 16-node adversarial tree, root `validated`, taint `16/16 clean`, closed in run 1. The export is `proofs/lem-negpart-subadditive/export.md`. The lemma was pre-factored *before* elevation, applying the balloon lesson from `lem-fan-payment` instead of re-learning it.
Role in the chain. Pre-factored dependency of 12: the D-restricted fan proof's barycenter step $D \geq n(\sum_{i \in A} p_i w_i)$ uses exactly this pointwise subadditivity (with nonnegative homogeneity, which is definitional).

13 D-restricted zero-sum fan payment

Lemma 12 (`lem-fan-payment-restricted`). *Let $(w_1, p_1), \ldots, (w_m, p_m)$ be a finite family with vectors $w_i \in \mathbb{R}^d$ and weights $p_i > 0$ satisfying $\sum_i p_i = 1$, such that every w_i has zero coordinate sum and the weighted barycenter is zero, $\sum_i p_i w_i = 0$. Write $n(w) = \sum_l \max(-w(l), 0)$, let w_* be a minimizer of $v \mapsto \sum_i p_i\, n(w_i - v)$ over $\{w_1, \ldots, w_m\}$, and set $A = \{i : n(w_i - w_*) > 0\}$. Then*

$$\sum_{i \in A} p_i\, n(w_i - w_*) \;\le\; (2 + \sqrt{2}) \sum_{i \in A} p_i\, n(w_i).$$

Registry contract (verbatim).

```
D-restricted zero-sum fan payment: let (w_1, p_1), ..., (w_m, p_m) be a finite family with vectors
w_i in R^d and weights p_i > 0 satisfying sum_i p_i = 1, such that every w_i has coordinate sum
zero and the weighted barycenter is zero (sum_i p_i w_i = 0); write n(w) = sum_l max(-w(l), 0); let
w_* be a minimizer of v -> sum_i p_i n(w_i - v) over {w_1, ..., w_m} and let A = { i : n(w_i - w_*)
> 0 }; then sum_{i in A} p_i n(w_i - w_*) <= (2 + sqrt(2)) * sum_{i in A} p_i n(w_i).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 27-node adversarial tree, root `validated`, taint `27/27 clean`, closed in run 1 after ten rounds; it imports the `af`-validated dependencies 8 and 11 as externals. The export is `proofs/lem-fan-payment-restricted/export.md`. Proof shape: with q the duplicate-pivot mass, horn (i) gives $N \le D/q$ and horn (ii) gives $N \le 2(1-q)/(1-2q)\,D$ for $q < 1/2$; the bounds cross at $q_0 = 1 - 1/\sqrt{2}$ with common value $2 + \sqrt{2}$.
Role in the chain. The D-restricted refinement of 10: the denominator keeps only rows with positive distance to the selected pivot, the fan shadow of the degenerate-set restriction behind `conj-degenerate-payment`. The registry shard body records exact certificates (wave material, not elevated here) refuting constant 2 for this variant and attaining $2+\sqrt{2}$ in the limit, so the validated constant is sharp. The remaining obstruction toward `conj-degenerate-payment` is the lift from θ-$\frac{1}{2}$ argmin rows to this fan model.

14 Pivot-removing max-stationarity

Lemma 13 (`lem-pivot-removing-move`). *Let P be a rank-3 exact signed idempotent ($P^2 = P$, all row sums 1), let $U = (u_0, u_1, u_2)$ be an actual-row chart whose rows $p_{u_0}, p_{u_1}, p_{u_2}$ form a basis of the row space, let $\mathrm{Vol}_{\max}(P)$ be the maximum Gram volume over actual-row charts, and $m_U = \mathrm{Vol}(U)/\mathrm{Vol}_{\max}(P)$. Define old coordinates $a_t(i)$ by $p_i = \sum_t a_t(i)\, p_{u_t}$, and set*

$$\beta_r(i) = P_{u_r i}, \qquad \lambda_r(i) = 1 - a_r(i), \qquad \mu_r(i) = \sum_{t \neq r} \max(-a_t(i), 0),$$

$$E_r(i) = \max(\mu_r(i) - \lambda_r(i), 0), \qquad \Phi_r(U) = \sum_i \max(\beta_r(i), 0)\, E_r(i), \qquad \Phi(U) = \max_r \Phi_r(U).$$

Assume U is a θ-$\frac{1}{2}$ Φ-argmin: $m_U \geq \frac{1}{2}$ and $\Phi(U)$ is minimal among actual-row charts W with $\mathrm{Vol}(W)/\mathrm{Vol}_{\max}(P) \geq \frac{1}{2}$. Fix a maximal pivot s with $\Phi_s(U) = \Phi(U)$, and let $j \notin \{u_0, u_1, u_2\}$ have $c = a_s(j) \neq 0$ such that the pivot-removing chart $V_j = U - u_s + j$ is θ-$\frac{1}{2}$ admissible—equivalently $|a_s(j)|\, m_U \geq \frac{1}{2}$, with volume factor $\mathrm{Vol}(V_j)/\mathrm{Vol}(U) = |a_s(j)|$. Defining new coordinates on V_j by

$$a^j_s(i) = \frac{a_s(i)}{c}, \qquad a^j_t(i) = a_t(i) - \frac{a_s(i)\, a_t(j)}{c} \quad (t \neq s),$$

$$E^j_r(i) = \max\Big(\sum_{q \neq r} \max(-a^j_q(i), 0) - (1 - a^j_r(i)),\ 0\Big),$$

$$\Psi_j = \Phi_s(V_j) = \sum_i \max(P_{ji}, 0)\, E^j_s(i), \qquad \Gamma_j = \max_{r \neq s} \Phi_r(V_j) = \max_{r \neq s} \sum_i \max(P_{u_r i}, 0)\, E^j_r(i),$$

one has

$$\Phi_s(U) \ \leq\ \max(\Psi_j, \Gamma_j).$$

Registry contract (verbatim).

```
Pivot-removing max-stationarity: let P be a rank-3 exact signed idempotent (square real matrix with
P^2 = P and all row sums equal to 1), let U = (u_0,u_1,u_2) be an actual-row chart whose rows
p_{u_0}, p_{u_1}, p_{u_2} form a basis of the row space, let Vol_max(P) be the maximum Gram volume
over actual-row charts, m_U = Vol(U)/Vol_max(P), define old coordinates a_t(i) by p_i = sum_t
a_t(i)p_{u_t}, beta_r(i) = P_{u_r i}, lambda_r(i) = 1-a_r(i), mu_r(i) = sum_{t != r}
max(-a_t(i),0), E_r(i) = max(mu_r(i)-lambda_r(i),0), Phi_r(U) = sum_i max(beta_r(i),0)E_r(i), and
Phi(U) = max_r Phi_r(U), assume U is a theta-half Phi-argmin meaning m_U >= 1/2 and Phi(U) is
minimal among actual-row charts W with Vol(W)/Vol_max(P) >= 1/2, fix a maximal pivot s with
Phi_s(U)=Phi(U), and let j notin {u_0,u_1,u_2} have c = a_s(j) != 0 such that the pivot-removing
chart V_j = U-u_s+j is theta-half admissible, equivalently |a_s(j)|*m_U >= 1/2 with volume factor
Vol(V_j)/Vol(U)=|a_s(j)|; defining new coordinates on V_j by a_s^j(i)=a_s(i)/c and
a_t^j(i)=a_t(i)-a_s(i)a_t(j)/c for t != s, E_r^j(i)=max(sum_{q != r}
max(-a_q^j(i),0)-(1-a_r^j(i)),0), Psi_j=Phi_s(V_j)=sum_i max(P_{ji},0)E_s^j(i), and Gamma_j=max_{r
!= s} Phi_r(V_j)=max_{r != s} sum_i max(P_{u_r i},0)E_r^j(i), one has Phi_s(U) <= max(Psi_j,
Gamma_j).
```

Status. `proved`; `af: validated`. This is one of the rigorous in-repo results.
Proof tree. The registry shard records a 9-node adversarial tree, root `validated`, taint `9/9 clean`, closed in run 1 after three rounds with zero challenges. The export is `proofs/lem-pivot-removing-move/export.r` The statement was codified from the arm-G wave record with a fully-inline contract and elevated pre-factored, so the tree stayed small.

Role in the chain. This is the minimality tool of the argmin engine: it is the first statement of the campaign that genuinely uses Φ-minimality of the chart, via the pivot-removing move with exact volume factor $|a_s(j)|$. Every attack on the remaining self-support control (`conj-sc` and its pivot-removing blocker reduction) consumes this disjunction: an admissible pivot-removing move must be blocked by the new-pivot score Ψ_j or by collateral Γ_j on another pivot. The wave-9 record realizes the volume-inadmissible and Ψ-blocked alternatives by exact certified instances; the collateral alternative is the open branch, and its analysis now rests on this validated statement.

15 The hiddenness dual witness and its always-tight support

Throughout this section P is an exact signed idempotent (`def-signed-idempotent`) with $\delta = \delta(P) > 0$, and we write the derived scales

$$\tau = \sqrt{\delta}, \qquad \rho = 4\tau, \qquad \kappa = \tfrac{1}{4}\tau,$$

as fixed in `CONVENTIONS.md` (c). For a row vertex v with row point p_v set $d_i = p_i - p_v$ and let

$$F_v = \{\, j : \|p_j - p_v\|_1 \geq \rho \,\}$$

be the ρ-far row-index set. Recall (`def-exposed`) that the *exposedness margin* of v is

$$t^*(v) = \sup_h \min_{f \in F_v} h(p_f), \tag{2}$$

the supremum over affine *admissible exposers* h with $h(p_v) = 0$ and $0 \leq h(p_i) \leq 1$ for every row i; the vertex v is *hidden* when it is not (ρ, κ)-exposed, i.e. $t^*(v) < \kappa$.

15.1 Existence of the witness

Lemma 14 (`lem-hiddenness-dual-witness`). *Let v be a hidden row vertex of P. Then $F_v \neq \varnothing$ and there exist coefficients $\lambda_f \geq 0$ ($f \in F_v$) with $\sum_f \lambda_f = 1$, and $\alpha_i, \beta_i \geq 0$ over all row indices with $\sum_i \beta_i = t^*(v) < \kappa$, such that*

$$\sum_{f \in F_v} \lambda_f\,(p_f - p_v) \;+\; \sum_i \alpha_i\,(p_i - p_v) \;=\; \sum_i \beta_i\,(p_i - p_v). \tag{3}$$

Registry contract (verbatim).

```
Hiddenness dual witness: for an exact signed idempotent P and a hidden row vertex v (rho = 4*tau,
kappa = tau/4, tau = sqrt(delta(P))), writing F_v = {j : ||p_j - p_v||_1 >= rho} for the rho-far
row-index set (nonempty for hidden v), there exist lambda_f >= 0 (f in F_v) with sum_f lambda_f = 1
and alpha_i, beta_i >= 0 (over all row indices i) with sum_i beta_i = t*(v) < kappa, such that
sum_f lambda_f*(p_f - p_v) + sum_i alpha_i*(p_i - p_v) = sum_i beta_i*(p_i - p_v).
```

Proof sketch. Hiddenness first supplies the scalar prerequisites: were F_v empty, `def-exposed` would give $t^*(v) = +\infty$, so a hidden vertex forces $F_v \neq \varnothing$ and $t^*(v) < \kappa$. Writing an admissible exposer as $h(x) = u \cdot (x - p_v)$, the margin (2) is the value of the finite linear program

$$\max_{u,\,t}\; t \quad \text{s.t.} \quad 0 \leq u \cdot d_i \leq 1 \;\; (\forall i), \qquad t \leq u \cdot d_f \;\; (\forall f \in F_v), \tag{4}$$

which is feasible at $(u, t) = (0, 0)$ and, since $F_v \neq \varnothing$, bounded above by 1. Dualising (4) with multipliers $\beta_i \geq 0$ on $u \cdot d_i \leq 1$, $\alpha_i \geq 0$ on $-u \cdot d_i \leq 0$, and $\lambda_f \geq 0$ on $t - u \cdot d_f \leq 0$ gives

$$\min \sum_i \beta_i \quad \text{s.t.} \quad \sum_f \lambda_f = 1, \qquad \sum_f \lambda_f d_f + \sum_i \alpha_i d_i = \sum_i \beta_i d_i.$$

Finite LP strong duality and dual attainment produce an optimal dual point with $\sum_i \beta_i = t^*(v)$; substituting $d_i = p_i - p_v$ is exactly (3), and $t^*(v) < \kappa$ closes the count. $\square$

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-hiddenness-dual-witness` (export node 1). First-principles finite LP duality with no imports; a fresh-codex hostile verifier re-derived feasibility, boundedness, attainment, and strong duality from scratch and checked the witness on the banked 7×7 duplicate-split fixture ($t^*(v) = \frac{1}{21} < \frac{1}{16}$).

15.2 The witness support is always tight

Fix a hidden geometrically distinct row vertex u with nonempty visible set $W(P)$. On the primal optimal face of (4) define the *always-tight* constraint families

$T = \{f \in F_u : h(p_f) = t^*(u) \,\forall \text{ optima } h\}$, $\quad O = \{i : h(p_i) = 1 \,\forall \text{ optima}\}$, $\quad Z = \{i : h(p_i) = 0 \,\forall \text{ optima}\}$.

Lemma 15 (`lem-always-tight-dual-support`)**.** *After deleting redundant centered-zero constraints, every optimal hiddenness dual witness (λ, α, β) of u satisfies*

$$\mathrm{supp}(\lambda) \subseteq T, \qquad \mathrm{supp}(\beta) \subseteq O, \qquad \mathrm{supp}(\alpha) \subseteq Z,$$

with $T \neq \varnothing$, and $O \neq \varnothing$ if and only if $t^*(u) > 0$.

Registry contract (verbatim).

```
Always-tight dual support: for the exposedness LP at a hidden geometrically distinct row vertex u
of an exact signed idempotent P with delta(P) > 0 and nonempty visible set, every optimal
hiddenness dual witness (lambda, alpha, beta), after deleting redundant centered-zero constraints,
has supp(lambda) contained in T, supp(beta) contained in O, and supp(alpha) contained in Z, where
T, O, Z are the rho-far, upper-box, and lower-box constraint families tight on the WHOLE primal
optimal face; T is nonempty, and O is nonempty if and only if t*(u) > 0.
```

Proof sketch. Pair any primal optimum h with the dual balance of Lemma 14. Using $h(p_u) = 0$, $\sum_f \lambda_f = 1$, and $\sum_i \beta_i = t^*(u)$ yields the *complementarity identity*

$$\sum_f \lambda_f\big(h(p_f) - t^*(u)\big) + \sum_i \alpha_i\, h(p_i) + \sum_i \beta_i\big(1 - h(p_i)\big) = 0. \tag{5}$$

Every summand of (5) is nonnegative by primal feasibility, so each vanishes: a positive λ_f forces $h(p_f) = t^*(u)$, a positive α_i forces $h(p_i) = 0$, and a positive β_i forces $h(p_i) = 1$. As h ranges over all optima this places the supports in T, Z, and O. Since $\sum_f \lambda_f = 1$ the set $\mathrm{supp}(\lambda)$ is nonempty, so $T \neq \varnothing$; and $O \neq \varnothing$ iff some $\beta_i > 0$ iff $\sum_i \beta_i = t^*(u) > 0$ (when $t^*(u) = 0$ the zero exposer is optimal and no upper box is tight). □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-always-tight-dual-support` (export node 1), importing Lemma 14 as a validated external. Two independent fresh-codex provers produced the same complementarity core; a separate hostile verifier added the redundant-centered-zero caveat (retained in the "reduced witness" clause) and the $t^* = 0$ edge.

Role in the chain. These two lemmas are the geometric engine of the hiddenness route: Lemma 14 converts "v is hidden" into a normalised transport identity on the ρ-far rows, and Lemma 15 pins that transport to the tight face. Their consumers are the depth-Markov concentration and far-row existence results of §16. Nothing here bounds the height H; that is the open cap of the Kernel Conjecture.

16 Consequences of the hiddenness witness

The scales $\tau = \sqrt{\delta}$, $\rho = 4\tau$, $\kappa = \frac{1}{4}\tau$, the far set F_v, and the exposedness margin $t^*(v)$ are those of §15. Write $C = \operatorname{conv}\{p_w : w \in W(P)\}$ and $d_j = \operatorname{dist}_1(p_j, C)$ for the depth of row j; a hidden top vertex v has height $H = d_v = \max_j d_j$ (`def-height`). The three results below all price the witness of Lemma 14 against an affine 1-Lipschitz functional, using the universal row-separation bound $\|p_i - p_k\|_1 \le (2 + 4\delta)$ of `def-signed-idempotent`.

16.1 Depth-Markov concentration

Lemma 16 (`lem-hiddenness-depth-markov`). *Let v be a hidden top vertex of height H and (λ, α, β) any hiddenness dual witness of v with $\sum_i \beta_i < \kappa$. Then for every $c > 0$,*

$$\lambda\big\{\, f \in F_v : d_f > H - c\,\tau \,\big\} \;>\; 1 - \frac{\frac{1}{2} + \delta}{c}. \tag{6}$$

Registry contract (verbatim).

```
Hiddenness depth-Markov: for an exact signed idempotent P with delta(P) > 0, nonempty visible set
W(P), hidden top vertex v of height H, and any hiddenness dual witness (lambda, alpha, beta) of v
with sum_i beta_i < kappa = tau/4 (tau = sqrt(delta)), one has for every c > 0: lambda{f in F_v :
dist_1(p_f, conv{p_w : w in W}) > H - c*tau} > 1 - (1/2 + delta)/c.
```

Proof sketch. Finite-dimensional ℓ_1 separation at p_v gives an affine functional ϕ with $\phi(p_v) = H$, linear part of ℓ_∞-norm at most 1, and $\phi(x) \le \operatorname{dist}_1(x, C)$; set $\psi = H - \phi$, so $\psi(p_v) = 0$. Then for every row i,

$$0 \le H - d_i \le \psi(p_i) \le \|p_v - p_i\|_1 \le (2 + 4\delta).$$

Applying the linear part of ψ to the witness balance (3) and using $\psi(p_v) = 0$, $\alpha_i, \beta_i \ge 0$, and $\sum_i \beta_i < \kappa$ gives

$$\sum_f \lambda_f\, \psi(p_f) \le \sum_i \beta_i\, \psi(p_i) < \kappa\,(2 + 4\delta) = \big(\tfrac{1}{2} + \delta\big)\tau,$$

whence $\sum_f \lambda_f (H - d_f) < (\frac{1}{2} + \delta)\tau$. Markov's inequality for the nonnegative random variable $H - d_f$ under the probability λ on F_v yields (6). □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-hiddenness-depth-markov` (export node 1), importing Lemma 14. The hostile verifier *weakened* the hypotheses: the c-parametric bound needs neither tallness nor a δ cutoff, only $\sum_i \beta_i < \kappa$, and it holds for *every* witness tuple.

16.2 A dual certificate for the margin, and a far top-depth row

Lemma 17 (`lem-row-far-dual-certificate`). *Let v be a geometrically distinct row vertex with $\delta > 0$, and set $L_F(v) = \sum_{f : \|p_f - p_v\|_1 \ge \rho} P^+_{vf}$ with $P^+_{vf} = \max(P_{vf}, 0)$ and ν_v the row-v negative mass. If $L_F(v) > 0$ then*

$$t^*(v) \;\le\; \frac{\nu_v}{L_F(v)}. \tag{7}$$

Registry contract (verbatim).

```
Row-far dual certificate: for an exact signed idempotent P and a geometrically distinct row vertex
v with delta(P) > 0 (tau = sqrt(delta), rho = 4*tau), writing L_F(v) = sum over {f : ||p_f -
p_v||_1 >= rho} of max(P_vf, 0) and nu_v the row negative mass, if L_F(v) > 0 then t*(v) <= nu_v /
L_F(v), where t*(v) is the exposedness margin of def-exposed.
```

Proof sketch. For any admissible exposer h, row reproduction $p_v = \sum_j P_{vj} p_j$ with $\sum_j P_{vj} = 1$ and $h(p_v) = 0$ give $0 = \sum_j P_{vj}\, h(p_j)$. Splitting into sign parts and using $0 \le h(p_j) \le 1$ bounds the positive far contribution:

$$\sum_{f \in F_v} P^+_{vf}\, h(p_f) \le \sum_{j:P_{vj}>0} P_{vj} h(p_j) = \sum_{j:P_{vj}<0} (-P_{vj}) h(p_j) \le \nu_v.$$

Dividing the weighted average by $L_F(v) > 0$ gives $\min_{f \in F_v} h(p_f) \le \nu_v / L_F(v)$ for every h; taking the supremum over h in (2) yields (7). □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-row-far-dual-certificate` (export node 1); first-principles weak duality (no attainment needed). Sharp on the banked W29 frontier instance ($\nu_3/L_F = \frac{1}{81} = t^*(3)$).

Lemma 18 (`lem-top-slab-companion`). *Let $0 < \delta \le \frac{1}{2}(17 - 12\sqrt{2})$, let $W(P)$ be nonempty, and let v be a hidden top vertex of height $H > 13\,\tau$. Then there is a row f with $\|p_f - p_v\|_1 \ge 4\tau$ and*

$$\operatorname{dist}_1\bigl(p_f, \operatorname{conv}\{p_w : w \in W\}\bigr) \;>\; H - \bigl(\tfrac{1}{2} + \delta\bigr)\tau. \tag{8}$$

Registry contract (verbatim).

```
Top-slab companion: for an exact signed idempotent P with 0 < delta(P) <= (17 - 12*sqrt(2))/2,
nonempty visible set W(P), and hidden top vertex v of height H > 13*tau (tau = sqrt(delta)), there
is a row f with ||p_f - p_v||_1 >= 4*tau and dist_1(p_f, conv{p_w : w in W}) > H - (1/2 +
delta)*tau.
```

Proof sketch. With the same $\psi = H - \phi$ as above ($0 \le \psi(p_i) \le (2 + 4\delta)$), Lemma 14 supplies a witness with $\sum_f \lambda_f \psi(p_f) \le \sum_i \beta_i \psi(p_i) < \kappa\,(2 + 4\delta) = (\frac{1}{2} + \delta)\tau$. Since the λ_f are a probability on F_v, some $f \in F_v$ has $\psi(p_f) < (\frac{1}{2} + \delta)\tau$; that f satisfies $\|p_f - p_v\|_1 \ge \rho = 4\tau$ and, via $\phi = H - \psi \le \operatorname{dist}_1(\cdot, C)$, inequality (8). The constant window $\delta \le \frac{1}{2}(17 - 12\sqrt{2})$ makes $13 - (\frac{1}{2} + \delta) \ge 4 + 6\sqrt{2} > 4$, so the far row is genuinely deep. □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-top-slab-companion` (export node 1), importing Lemma 14. The verifier checked the exact symbolic constant chain.

Role in the chain. Lemma 16 shows the witness mass concentrates within an $O(\tau)$ slab below the top, Lemma 17 turns a positive far mass into a quantitative margin bound, and Lemma 18 extracts a single deep far row. Together they are the witness-side inputs to the halo-collapse height cap; the cap itself (1) stays open.

17 Slab and capacity primitives

The three primitives below are elementary consequences of exact idempotence. For an exact signed idempotent P (`def-signed-idempotent`) each row satisfies the *reproduction identity*

$$p_v = \sum_j P_{vj}\, p_j, \qquad \sum_j P_{vj} = 1, \tag{9}$$

so for any affine h with $h(p_v) = 0$ one has $0 = \sum_j P_{vj}\, h(p_j)$. Write $P^+_{vj} = \max(P_{vj}, 0)$, $P^-_{vj} = \max(-P_{vj}, 0)$, and $\nu_v = \sum_j P^-_{vj}$ for the row-v negative mass (`def-negative-mass`).

17.1 The Chebyshev low-slab pincer

Lemma 19 (`lem-cs-low-slab-pincer`). *Let v be any row index, let h be affine with $h(p_v) = 0$ and $0 \le h(p_j) \le 1$ for every row j, and let $s > 0$. Then*

$$\sum_{j:\, h(p_j) \ge s} P^+_{vj} \;\le\; \frac{\nu_v}{s}. \tag{10}$$

Registry contract (verbatim).

```
CS low-slab pincer: for an exact signed idempotent P, a row index v with nu_v = sum_j max(-P_vj,
0), an affine h with h(p_v) = 0 and 0 <= h(p_j) <= 1 for every row j, and every s > 0, one has sum
over {j : h(p_j) >= s} of max(P_vj, 0) <= nu_v / s.
```

Proof sketch. From (9) and $h(p_v) = 0$, the sign split gives $\sum_j P^+_{vj} h(p_j) = \sum_j P^-_{vj} h(p_j) \le \sum_j P^-_{vj} = \nu_v$, using $0 \le h(p_j) \le 1$. On the slab $A_s = \{j : h(p_j) \ge s\}$, $s\, P^+_{vj} \le P^+_{vj} h(p_j)$, so $s \sum_{A_s} P^+_{vj} \le \sum_j P^+_{vj} h(p_j) \le \nu_v$; divide by $s > 0$. □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-cs-low-slab-pincer` (export node 1). The verifier confirmed the minimal hypotheses (no hiddenness, no vertex, no $\delta > 0$) and that equality is attained at $s = t^*$ on the W29 frontier instance, so (10) is sharp.

17.2 Row-zero capacity

Lemma 20 (`lem-row-zero-capacity`). *Let i be a row index and y a vector with $Py = y$, $0 \le y_j \le 1$ for all j, $y_i = 0$, and $y_f \ge \kappa$ for all f in a set F. Then*

$$\kappa \sum_{f \in F} P^+_{if} \;\le\; \nu_i. \tag{11}$$

Registry contract (verbatim).

```
Row-zero capacity: for an exact signed idempotent P, a row index i, and any vector y with P y = y,
0 <= y_j <= 1 for all j, y_i = 0, and y_f >= kappa for all f in a set F, one has kappa * sum over f
in F of max(P_if, 0) <= nu_i, where nu_i is the row-i negative mass.
```

Proof sketch. The i-th coordinate of $Py = y$ with $y_i = 0$ gives $0 = \sum_j P_{ij} y_j$, hence $\sum_j P^+_{ij} y_j = \sum_j P^-_{ij} y_j \le \sum_j P^-_{ij} = \nu_i$ by $0 \le y_j \le 1$. On F, $y_f \ge \kappa$ gives $\kappa \sum_{f\in F} P^+_{if} \le \sum_{f \in F} P^+_{if} y_f \le \sum_j P^+_{ij} y_j \le \nu_i$. □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-row-zero-capacity` (export node 1), resting on Lemma 21 in the registry (which certifies the $Py = y$ hypothesis is the harmonic condition). The verifier made the $y_i = 0$ hypothesis explicit.

17.3 The harmonic–affine bridge

Lemma 21 (`lem-harmonic-affine-bridge`). *For an exact signed idempotent P with rows $p_i = (P_{ij})_j$, a vector g satisfies $Pg = g$ if and only if there is a vector u with $g_i = u \cdot p_i$ for every row index i.*

Registry contract (verbatim).

```
Harmonic-affine bridge: for an exact signed idempotent P with rows p_i = (P_ij)_j, a vector g
satisfies Pg = g if and only if there exists u with g_i = u . p_i for every row index i; in the
forward direction u = g works (g_i = p_i . g), and the constant term of any affine representation
is absorbable into u since all row sums equal 1.
```

Proof sketch. ($\Rightarrow$) If $Pg = g$ then $g_i = (Pg)_i = p_i \cdot g$, so $u = g$ works. ($\Leftarrow$) If $g_i = u \cdot p_i$ then by row reproduction (9), $(Pg)_i = \sum_j P_{ij}(u \cdot p_j) = u \cdot \left(\sum_j P_{ij} p_j\right) = u \cdot p_i = g_i$. A constant term c in an affine representation $g_i = c + u \cdot p_i$ is absorbed by $u' = u + c\mathbf{1}$, since every row sum is 1. $\square$

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-harmonic-affine-bridge` (export node 1); two lines each direction, checked entrywise on the banked W19 rank-5 fixture.

Role in the chain. Lemma 21 identifies the P-harmonic vectors ($Pg = g$) with affine test functions of the rows, so an admissible exposer is exactly a box-constrained harmonic vector; Lemmas 19 and 20 are then the two capacity estimates that price positive far mass against ν_v. They feed the same halo-collapse and starvation arguments as the witness lemmas of §16.

18 Parametric and depth-d halo collapse

Both results generalise the af-validated halo-collapse bridge 6 by making the halo width a free parameter. Fix an exact signed idempotent P with $0 < \delta \le \frac{1}{4}$ and nonempty visible set $W(P)$; write $C = \mathrm{conv}\{p_w : w \in W\}$, $d_j = \mathrm{dist}_1(p_j, C)$, $\tau = \sqrt{\delta}$, and $\alpha_j = P_{vj}$ with parts α_j^+, α_j^- and $\nu_v = \sum_j \alpha_j^-$. For a halo width $a > 0$ the mass placed strictly outside the $a\tau$-halo of C is $\sigma_a = \sum_{j:\, d_j > a\tau} \alpha_j^+$, and $\sigma = \sum_{j:\, d_j > 0} \alpha_j^+$ is the invisible mass (`def-invisible-mass`).

18.1 The parametric bridge

Lemma 22 (`lem-parametric-halo-collapse`). *Let v be a hidden top vertex of height H and $a > 0$. Then*

$$H\,(1-\sigma_a) \;\le\; (\sigma - \sigma_a)\,a\tau \;+\; \nu_v\,(2+4\delta). \tag{12}$$

Registry contract (verbatim).

```
Parametric halo collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty
visible set W(P), hidden top vertex v of height H, and any halo width a > 0, writing sigma_a for
the positive coefficient mass v places on rows at ell-1 distance > a*tau from conv W (tau =
sqrt(delta)), sigma for the invisible mass, and nu_v for the row negative mass, one has H*(1 -
sigma_a) <= (sigma - sigma_a)*a*tau + nu_v*(2 + 4*delta).
```

Proof sketch. If $\sigma_a \ge 1$ the left side is ≤ 0 and the right side is ≥ 0. Otherwise put $A = \{j : d_j > a\tau\}$ and

$$q = \frac{1}{1-\sigma_a}\Big(\sum_{j\notin A} \alpha_j^+\, p_j - \sum_k \alpha_k^-\, p_k\Big),$$

so that by row reproduction and mass split ($\sum_j \alpha_j^+ = 1 + \nu_v$, 3) $p_v = \sum_{j\in A} \alpha_j^+\, p_j + (1-\sigma_a)\, q$ is a genuine convex combination. The lower residual estimate 4 gives $H \le \mathrm{dist}_1(q, C)$, while the upper residual estimate 5 with $D_k = (2+4\delta)$ gives $(1-\sigma_a)\,\mathrm{dist}_1(q, C) \le \sum_{j\notin A} \alpha_j^+ d_j + \nu_v(2+4\delta)$. Pricing the non-halo rows by $\sum_{j\notin A} \alpha_j^+ d_j \le (\sigma - \sigma_a)a\tau$ and multiplying the first bound by $1-\sigma_a > 0$ yields (12). ☐

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-parametric-halo-collapse` (export node 1), importing 3, 4, 5. At $a = \frac{1}{4}$ it recovers the 6 contract verbatim; the verifier checked that calibration and the rank-5 fixture.

18.2 The depth-d version

Lemma 23 (`lem-depth-d-halo-collapse`). *Let v be any row index with depth $d = \mathrm{dist}_1(p_v, C)$ and let $a > 0$. With $\sigma_a(v) = \sum_{d_j > a\tau} P_{vj}^+$, $\sigma(v) = \sum_{d_j > 0} P_{vj}^+$, and the deeper-row correction $C_a(v) = \sum_{d_j > a\tau,\ d_j > d} P_{vj}^+\,(d_j - d)$,*

$$d\,(1-\sigma_a(v)) \;\le\; (\sigma(v) - \sigma_a(v))\,a\tau \;+\; \nu_v\,(2+4\delta) \;+\; C_a(v). \tag{13}$$

Registry contract (verbatim).

```
Depth-d halo collapse: for an exact signed idempotent P with 0 < delta(P) <= 1/4 and nonempty
visible set W(P), any row index v with d = dist_1(p_v, conv{p_w : w in W}), and any halo width a >
0, writing d_j = dist_1(p_j, conv W), sigma_a(v) = sum over {j : d_j > a*tau} of max(P_vj, 0) (tau
```

```
= sqrt(delta)), sigma(v) = sum over {j : d_j > 0} of max(P_vj, 0), nu_v the row negative mass, and
C_a(v) = sum over {j : d_j > a*tau, d_j > d} of max(P_vj, 0)*(d_j - d), one has d*(1 - sigma_a(v))
<= (sigma(v) - sigma_a(v))*a*tau + nu_v*(2 + 4*delta) + C_a(v).
```

Proof sketch. The trivial branch $\sigma_a(v) \geq 1$ is as before. Otherwise the same residual q (now with $B = \{j : d_j \leq a\tau\}$) gives $p_v = \sum_{j\in A} P^+_{vj} p_j + (1-\sigma_a)q$. Convexity of $x \mapsto \mathrm{dist}_1(x, C)$ gives $d \leq \sum_{j\in A} P^+_{vj} d_j + (1-\sigma_a)\,\mathrm{dist}_1(q, C)$; splitting A at $d_j \leq d$ versus $d_j > d$ replaces $\sum_{j\in A} P^+_{vj} d_j$ by $\sigma_a(v)\, d + C_a(v)$, so $(1-\sigma_a)d \leq C_a(v) + (1-\sigma_a)\,\mathrm{dist}_1(q, C)$. Pricing $\mathrm{dist}_1(q, C)$ through 5 exactly as in the parametric case gives (13). At a hidden top ($d = H$, $C_a = 0$) it recovers (12). □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-depth-d-halo-collapse` (export node 1), importing 3, 5. The verifier legitimised the convex split and checked the top-row calibration against 22.

Role in the chain. These are the halo-collapse bounds at arbitrary width and depth: the free parameter a is what lets the genuine-mass bootstrap of §19 choose a halo that separates hidden from visible rows. They tighten but do not close the height cap of the Kernel Conjecture 1.

19 Genuine-mass disintegration and top concentration

Keep the notation of §18: $C = \operatorname{conv}\{p_w : w \in W(P)\}$, $d_j = \operatorname{dist}_1(p_j, C)$, $\tau = \sqrt{\delta}$, height H, and $P_{ij}^+ = \max(P_{ij}, 0)$. Fix a halo width $a > 0$ and the genuine (deep-halo) index set $G_a = \{j : d_j > a\tau\}$, assumed nonempty. The two results below are the "bootstrap" step: they show that the mass a row sends into G_a is either supported on genuinely hidden vertices or is small.

19.1 Disintegration of the halo mass

Lemma 24 (`lem-genuine-disintegration`). *Fix for every row j a convex vertex representation $p_j = \sum_v \lambda_{jv} p_v$ over geometrically distinct row vertices, and let $g = P\mathbf{1}_{G_a}$. Then every row index i satisfies*

$$g_i \;\le\; M_i^a \;+\; \sum_{j\in G_a} P_{ij}^+ \,\frac{H - d_j}{H - a\tau}, \qquad M_i^a = \sum_{j\in G_a} P_{ij}^+ \sum_{v:\, d_v > a\tau} \lambda_{jv}, \tag{14}$$

where $M_i^a \ge 0$ is positive mass supported entirely on hidden row vertices of depth in $(a\tau, H]$.

Registry contract (verbatim).

```
Genuine-mass disintegration: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty
visible set W(P), height H, halo width a > 0 with G_a = {j : dist_1(p_j, conv W) > a*tau} nonempty
(tau = sqrt(delta)), fix for every row a vertex representation p_j = sum_v lambda_jv p_v over
geometrically distinct row vertices; then every row index i satisfies g_i <= M_i^a + sum_{j in G_a}
P_ij^+ * (H - d_j)/(H - a*tau), where g = P*1_{G_a}, d_j = dist_1(p_j, conv W), P_ij^+ = max(P_ij,
0), and M_i^a = sum_{j in G_a} P_ij^+ * sum_{v : d_v > a*tau} lambda_jv is positive mass supported
entirely on HIDDEN row vertices at depth in (a*tau, H].
```

Proof sketch. Nonemptiness of G_a forces $H > a\tau$. Applying the upper residual estimate 5 to the vertex representation gives the depth-convexity bound $d_j \le \sum_v \lambda_{jv} d_v$. Splitting each $j \in G_a$ into shallow mass $L_j = \sum_{d_v \le a\tau} \lambda_{jv}$ and deep mass $U_j = 1 - L_j$ and using $d_v \le a\tau$ on the first class, $d_v \le H$ on the second, yields $d_j \le a\tau\, L_j + H\, U_j$, hence $L_j \le (H - d_j)/(H - a\tau)$. Since $g_i = \sum_{j\in G_a} P_{ij} \le \sum_{j\in G_a} P_{ij}^+$ and $1 = L_j + U_j$, weighting the L_j-bound by $P_{ij}^+ \ge 0$ gives (14); the U_j-part is M_i^a, whose vertices have $d_v > a\tau$ and hence are hidden of depth $(a\tau, H]$. □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-genuine-disintegration` (export node 1), importing 5. The verifier checked the classification, extreme-point identification, and denominators on the banked rank-3 and rank-5 instances (step 3 of sketch M1).

19.2 Top concentration

Lemma 25 (`lem-top-concentration`). *Let v be a hidden top vertex of height H and $a > 0$ with $H > a\tau$. The positive mass v places outside the genuine set G_a satisfies*

$$\sum_{j\notin G_a} P_{vj}^+ \;\le\; \frac{\nu_v\,(2 + 4\delta)}{H - a\tau}. \tag{15}$$

Registry contract (verbatim).

```
Top concentration: for an exact signed idempotent P with 0 < delta(P) <= 1/4, nonempty visible set
W(P), hidden top vertex v of height H, and halo width a > 0 with H > a*tau (tau = sqrt(delta)), the
positive mass v places outside the genuine set satisfies sum_{j notin G_a} P_vj^+ <=
```

```
nu_v*(2+4*delta)/(H - a*tau), where G_a = {j : dist_1(p_j, conv{p_w : w in W}) > a*tau} and nu_v is
the row-v negative mass.
```

Proof sketch. Choose an ℓ_1/ℓ_∞ support functional $\phi(x) = u \cdot x + b$ with $\|u\|_\infty \leq 1$, $\phi \leq 0$ on C, and $\phi(p_v) = H$; set $t_j = H - \phi(p_j)$. Then $0 \leq t_j \leq (2+4\delta)$ (row separation), $t_j \geq H - d_j \geq H - a\tau$ for $j \notin G_a$, and by row reproduction $\sum_j P_{vj} t_j = 0$. The sign split gives $\sum_j P^+_{vj} t_j = \sum_j P^-_{vj} t_j \leq (2+4\delta) \sum_j P^-_{vj} = \nu_v(2+4\delta)$. Restricting to $j \notin G_a$, $(H - a\tau) \sum_{j \notin G_a} P^+_{vj} \leq \sum_j P^+_{vj} t_j \leq \nu_v(2+4\delta)$; divide by $H - a\tau > 0$. □

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-top-concentration` (export node 1); first-principles support functional, no imports. The verifier ran an exact LP support-functional test on the banked rank-5 instance (step 4 decider).

Role in the chain. Lemma 25 shows a hidden top vertex sends all but an $O(\nu_v/H)$ fraction of its positive mass into the genuine set G_a, and Lemma 24 then re-expresses that genuine mass as mass on strictly hidden vertices plus a controlled shallow remainder. Iterating these against the depth-d collapse 23 is the g-bootstrap intended to force $H = O(\tau)$; the closing cap remains the open Kernel Conjecture 1.

20 The bounded-slab starvation completion obstruction

This standalone obstruction rules out a rank-three completion of the exchange-starvation configuration without any auxiliary constant K. Fix a finite index set I, a real $A \in [4,6]$, a real $\tau \in (0, \frac{1}{256}]$, and set

$$t = \tau^2, \qquad a = \frac{\tau}{1+\tau}.$$

Lemma 26 (`lem-starvation-completion-obstruction`)**.** *There is no rank-three exact signed idempotent P (so $P^2 = P$, $P\mathbf{1} = \mathbf{1}$, $\operatorname{rank} P = 3$) with row negative mass $\nu_i \le t$ for every $i \in I$ that carries five distinct full row-point fibers, represented by v, w, f, z, o, such that, writing $D = p_z - p_v$ and $E = p_o - p_v$ (linearly independent),*

$$\|D\|_1 = \tau, \quad p_f - p_v = -A\,D + t\,E, \quad p_w - p_v = a\,(p_f - p_v),$$

with top-row fiber masses $c_v = 1 - \tau$, $c_w = \tau + t$, $c_f = -t$, and $c_Q = 0$ for every other fiber Q, where every full row-point fiber has unique coordinates $p_Q = p_v + x_Q D + y_Q E$ and every nonactor support fiber satisfies $p_Q \in \operatorname{conv}\{p_v, p_w, p_f, p_z, p_o\}$ or $0 \le y_Q \le 1$.

Registry contract (verbatim).

```
Bounded-slab starvation completion obstruction (K-free): for every finite I, every real A in [4,6],
every real tau in (0,1/256], and t := tau^2, a := tau/(1+tau), there is no rank-three exact signed
idempotent P (P^2 = P, P*1 = 1, rank P = 3) with row negative mass nu_i <= t for every i in I
having five distinct full row-point fibers represented by v,w,f,z,o such that, with D := p_z - p_v
and E := p_o - p_v linearly independent, ||D||_1 = tau, p_f - p_v = -A*D + t*E, p_w - p_v = a*(p_f
- p_v), top-row fiber masses c_v = 1 - tau, c_w = tau + t, c_f = -t, and c_Q = 0 for every other
full row-point fiber Q, every full row-point fiber Q has unique reals x_Q, y_Q with p_Q = p_v +
x_Q*D + y_Q*E, and every nonactor support fiber Q satisfies either p_Q in conv{p_v,p_w,p_f,p_z,p_o}
or 0 <= y_Q <= 1.
```

Proof sketch (by contradiction). Suppose such a P exists. Group the coordinate indices into fibers Q (classes of equal rows) and set $c_Q = \sum_{j\in Q} P_{vj}$, $d_Q = \sum_{j\in Q}(P_{zj} - P_{vj})$, $e_Q = \sum_{j\in Q}(P_{oj} - P_{vj})$. The negative-mass hypothesis $\nu_i \le t$ gives the subset ledger $-t \le P_i(S) \le 1 + t$ and $\|p_i\|_1 \le 1 + 2t$ for every row i and subset S. Subtracting the v-row from the z-row of $P^2 = P$ gives $DP = D$ with $\sum_j D_j = 0$; expanding $D = \sum_j D_j p_j$ in the coordinates $p_j = p_v + x_j D + y_j E$ and using linear independence of D, E produces the *exact unit moment*

$$\sum_Q x_Q\, d_Q = 1. \tag{16}$$

Every actor and interior fiber has bounded x-coordinate, $|x_Q| \le A$, so the hull part contributes at most $A\tau$ (since $\sum_{Q\in \mathrm{Hin}} |d_Q| \le \|D\|_1 = \tau$). For an exterior fiber Q the disjunction forces $0 \le y_Q \le 1$, and the coordinate identity with the row-norm bounds gives the lever estimate $|x_Q|\tau \le 2 + 4t$; a single global budget from the row-z, row-f, and row-o subset bounds gives $\sum_{Q\in\mathrm{Ext}} |d_Q| \le t\bigl(1 + (2+t)/A\bigr)$, with *no* dependence on the number of fibers. Combining,

$$1 = \sum_Q x_Q d_Q \;\le\; \tau\Big[A + (2+4t)\big(1 + \tfrac{2+t}{A}\big)\Big].$$

But $0 < \tau \le \frac{1}{256}$ gives $t < \frac{1}{4}$, $2 + 4t < 3$, $(2+t)/A \le 3/4$, and $A \le 6$, so the right side is strictly below $\frac{1}{256}\bigl[6 + 3(1+\frac{3}{4})\bigr] = \frac{45}{1024} < 1$ — contradicting (16). ☐

Provenance. `status: proved`, `af: validated`; workspace `proofs/lem-starvation-completion-obstructi` (export node 1). W59 wave: a codex prover paper proof from first principles (guided by, but independent of, the W57/W58 exact Farkas certificates); a fresh hostile codex verifier returned `VALID-WITH-CORRECTIONS` (a single index-level notation correction applied). Reviewer $\neq$ author (`owner: B`).

Role in the chain. This is the "K-free" closure of the exchange-starvation leaf: the earlier reductions of that branch depended on an auxiliary completion constant, and this obstruction removes it by showing the rank-three completion cannot exist in the bounded-slab regime. It is the verified four-leaf-system input recorded in the live proof sketch; it does not by itself settle `op-classical`, which remains open.

21 Status ledger for non-validated registry results

The rows below are anchors, not upgrades. They give every remaining registry result a report label while preserving the registry status. Of the twenty-nine `af`-validated registry results, twenty-six are reproduced as their own sections (enumerated in the roadmap, §1) and are not duplicated in this ledger. The three validated results not yet given a dedicated section (`lem-collateral-import`, `lem-cross-pivot-cancellation`, `lem-import-reduction`) are anchored below at their true `af`-validated status.

Table 1: Registry status ledger for report anchors outside the thirteen validated sections.

Registry id	Status	Contract gloss
`conj-degenerate-payment`	conjecture	Pivot-local weighted near-degenerate payment bound; the payment horn of the `(EX)` argmin charge, exact-tested on the wave zoo but unproved.
`conj-degenerate-transport`	conjecture	Degenerate transverse transport bound; the remaining payment-horn gap after the wave-12 lift, exact-tested but unproved.
`conj-ex`	conjecture	`(EX)` chart-bound working form for exact signed idempotents; still a conjectural input equivalent to `conj-kernel` in the registry.
`conj-kernel`	conjecture	Kernel Conjecture, the geometric input intended to close `op-exposed-hull` and hence `op-classical`.
`conj-no-free-frontier`	conjecture	Exposedness-absorption statement from the frontier analysis; retained as conjectural and not elevated.
`conj-rh`	conjecture	Repaired orphan horn: post-fan orphan demand bounded by the unified class/signed/own-negativity budget plus fan residual; exact floor four on the coefficient.
`conj-sc`	conjecture	Argmin self-support/cancellation control for non-fan beta-positive rows; reduced to the pivot-removing blocker theorem, collateral branch open.
`conj-skinny-shadow-cap`	conjecture	Skinny two-shadow cap at the root-delta scale, the corrected Route-B statement superseding the retired dual-localization contract.
`ex-hume`	proved-mod-audit	Explicit 3-by-3 sharpness family for the exponent in `op-classical`; inherited proof is not rigorous here.
`lem-canonical-separator`	proved-mod-audit	Canonical separator package for a hidden top vertex and its harmonic deficit; inherited/mod-audit only.
`lem-dual-localization`	obstruction (superseded)	Retired 2026-07-04: the transcribed contract is a distance tautology; superseded by `conj-skinny-shadow-cap`.
`lem-exposed-circuit`	proved-mod-audit	Concentration and circuit estimate for exposed vertices; inherited/mod-audit only.
`lem-collateral-import`	proved (af: validated)	Collateral-import bound (CI) for the pivot-removing move, sharp on the wave-10 witness; af-validated in-repo 2026-07-04 (32-node tree, taint clean).
`lem-cross-pivot-cancellation`	proved (af: validated)	Cross-pivot B-L cancellation identity behind the import anatomy; af-validated in-repo 2026-07-04 (23-node tree, taint clean); near-definitional.

`lem-import-reduction`	proved (af: validated)	Universal reduction of the collateral import to the cross-pivot masses; af-validated in-repo (G11 reduction (4)); section shard pending.
`lem-leakage`	proved-mod-audit	Affine-face leakage estimate for almost-idempotent stochastic rows; inherited/mod-audit only.
`lem-wiggle-rigidity`	proved-mod-audit	F-WR wiggle-rigidity estimate for common-pattern webs; side conditions remain inherited.
`obs-deep-leakage`	heuristic	Heuristic obstruction showing why a purely local row-identity argument cannot price deep-side mass.
`obs-fwr-gap`	heuristic	F-WR forbidden-gap obstruction to using web rigidity as the dimension-free shallow-class counter.
`obs-linear-law-finite-delta`	numerical	Exact finite-delta certificate exceeding the empirical linear-law constant; L3 evidence, not proof.
`obs-orphan-amplifier`	proved-mod-audit	Exact two-orphan family killing every class/signed-only orphan budget; repaired ratio tends to four; wave paper-proof, orchestrator-recomputed.
`obs-sigma-halo-nonrobust`	numerical	Exact certificate that the literal zero-halo sigma cap is false; L3 evidence and obstruction.
`op-classical`	open	North-star classical projection stability problem; still open in this repo.
`op-exposed-hull`	open	Global exposed-hull lemma that would imply `op-classical` through the inherited factorization route; open.
`prop-approx-simplex`	proved-mod-audit	Approximate-simplex reduction to a stochastic idempotent; inherited/mod-audit only.
`thm-classical-factorization`	proved-mod-audit	Conditional classical factorization theorem through cluster geometry and `op-exposed-hull`; not rigorous here.
`thm-cluster`	proved-mod-audit	Cluster geometry reduction for signed affine retractions; inherited/mod-audit only.
`thm-corner-constants`	proved-mod-audit	Corner-scale constants, with numerical components explicitly not promoted.
`thm-rank-one`	proved-mod-audit	Rank-one signed affine retraction stability, including the Hume family; inherited/mod-audit only.
`thm-simplex`	proved-mod-audit	Simplex and one-dimensional row-polytope stability reduction; inherited/mod-audit only.
`thm-well-exposed`	proved-mod-audit	Well-exposed-vertex route into the simplex theorem; inherited/mod-audit only.